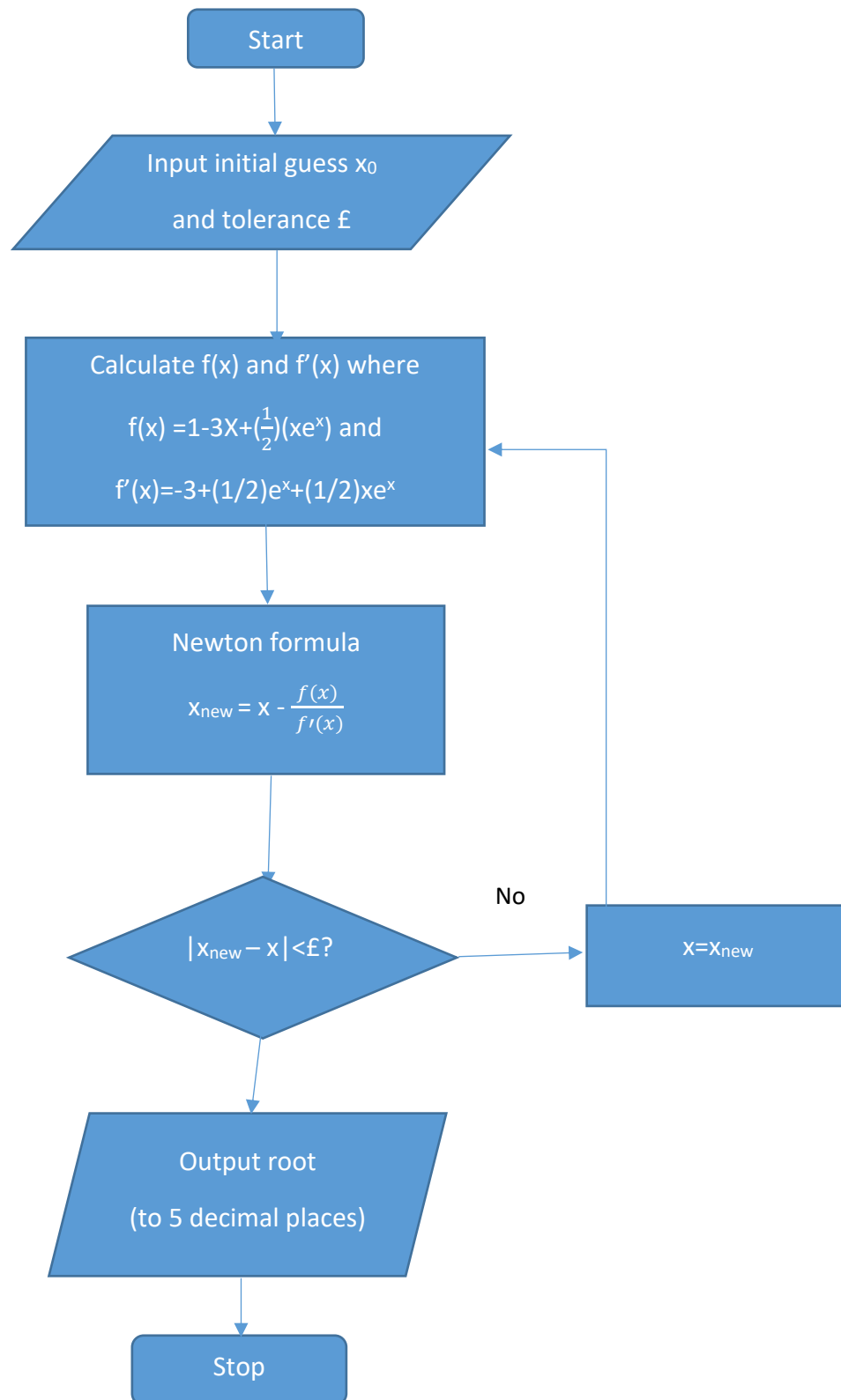


Use Newton's method to find the zeros of $f(x) = 1 - 3x + \frac{1}{2}(xe^x)$ accurate to five decimal places.



PSEUDOCODE

START

Define $f(x) = 1 - 3x + \left(\frac{1}{2}\right)xe^x$

Define $f'(x) = -3 + \left(\frac{1}{2}\right)e^x(1+x)$

Input initial guess x_0

Input tolerance ϵ

REPEAT

 Calculate $f(X_0)$

 Calculate $f'(X_0)$

$$X_1 = X_0 - \frac{f(x_0)}{f'(x_0)}$$

 IF $= |x_1 - x_0| < \epsilon$

 STOP

 ELSE

$$X_0 \leftarrow X_1$$

 END IF

UNTIL convergence

Display x_1 to 5 decimal places

STOP

'FOR' AND 'WHILE' LOOPS

```
f=@(x) 1-3*x + 0.5*x*exp(x);
df=@(x)-3+0.5*exp(x)*(1+x);
x0 = 0;
tol=0.000001;
N=100;
for i = 1:N
    x1 = x0-f(x0)/df(x0);
    if abs(x1-x0)<tol
        end
        x0=x1;
end
fprintf('Root=%.5f/n',x1);
fprintf('Iterations =%d/n',i);

while i<N
    x1 =x0-f(x0)/df(x0);
    i=i+1;
    if abs(x1-x0)<tol
        end
        x0=x1;
end
fprintf('Root=%.5f/n',x1);
fprintf('Iterations =%d/n',i);
```

MATLAB CODE

```
f=@(x)1-3*x +0.5*x.*exp(x);
df =@(x)-3+0.5*exp(x).*(1+x);
x0=0;
tol=1e-6;
maxiter=100;
for i=1:maxiter
    x1=x0-f(x0)/df(x0);
    if abs(x1-x0)<tol
        break;
    end
    x0=x1;
end
fprintf('Root=%.5f/n',x1);
fprintf('Number of iterations=%d/n',i);
```

